Skin Sensitization AOP 40 — Mechanistic Detail Extracted from a Decade of Focused Research

Source under review: Aleksic M, Rajagopal R, de-Ávila R, Spriggs S, Gilmour N. "The skin sensitization adverse outcome pathway: exploring the role of mechanistic understanding for higher tier risk assessment." *Crit Rev Toxicol* 2024. PMID: 38385441, DOI:10.1080/10408444.2024.2308816.

AOP-Wiki pathway: AOP 40, "Covalent Protein binding leading to Skin Sensitisation" (KE396 → KE826 → KE398 → KE272 → KE827).

Summary

The body of mechanistic work synthesized by this review establishes skin sensitization not as a qualitative on/off cascade but as a **quantitative, kinetics-anchored causal chain** in which each step is a specific molecular cause producing a specific consequence. The molecular initiating event (MIE/KE1) is covalent haptenation — formation of a stable adduct between a reactive electrophile and protein nucleophiles, preferentially cysteine thiols, and this modification is the step that creates the chemical-specific antigen recognized by T cells. Crucially, recent chemistry shows that KE1 is **rate-limiting**: sensitizer potency tracks the *rate* of cysteine-adduct formation (log kmax), not a binary "binds / does not bind" distinction. The downstream steps are keratinocyte stress and danger signaling (KE2: KEAP1–NRF2–ARE plus the NLRP3/ASC/caspase-1 inflammasome driving IL-1β/IL-18), dendritic cell maturation (KE3: p38-MAPK-dependent CD80/CD83/CD86 upregulation and MHC class II internalization, with AhR acting as a brake), hapten-specific T-cell proliferation (KE4: MHC class I presentation of hapten-modified neoantigens to polyclonal $CD8^+$ Tc1 effectors, restrained by $CD4^+$/$FoxP3^+$ regulatory T cells), and finally the adverse outcome (AO/KE827: a threshold-bearing, sigmoid dose–response governed by **dose per unit area, µg/cm²**).

The review's central interest — where recent mechanism **revises** the AOP as originally written — is supported by three well-evidenced revisions. First, Langerhans cells (LCs) are **not the obligatory antigen-presenting cell**; genetic ablation experiments show contact hypersensitivity is *amplified* rather than abolished when LCs are absent, so LCs are dispensable and even regulatory, and the identity of the priming dendritic-cell subset that replaces them remains unresolved. Second, metal sensitizers (Ni^{2+}, Co^{2+}) initiate sensitization **non-covalently**, by coordinating the non-conserved histidines H456/H458 of human TLR4 — a route that lies *outside* AOP 40's covalent MIE and is human-specific (mouse TLR4 cannot respond). Third, pre-haptens (abiotic oxidation) and pro-haptens (keratinocyte CYP450/NQO1 metabolism) generate the reactive electrophile upstream of KE1 rather than bypassing it, so they remain *inside* AOP 40 but shift where the reactive species is formed.

Interindividual variation has several demonstrated mechanisms that largely act by **tuning the effective hapten dose** reaching viable epidermis: filaggrin loss-of-function increases chemical penetration (nickel-specific association), N-acetyltransferase polymorphisms modify pro-hapten (p-phenylenediamine) detoxification, the skin microbiome amplifies innate responses to metals, and $CD25^{+}$ regulatory T cells actively maintain tolerance to nickel in healthy people. By contrast, HLA/MHC associations for *defined contact haptens* are weak and contradictory — an association at best, not a demonstrated restriction mechanism. Three steps remain mechanistically unresolved: the specific in-vivo epidermal protein target whose modification actually drives KE2/KE3 (current assays measure bulk cysteine-peptide depletion as a surrogate), the identity of the human priming APC subset, and a reproducible HLA basis for contact-hapten susceptibility.

Key Findings

KE1 / MIE — Covalent protein binding is thiol-dependent and rate-limiting

Haptenation is thiol-dependent adduct formation that creates the T-cell antigen. Working *in chemico* with the strong human sensitizer DNCB and its water-soluble form DNBS reacting against an all-amino-acid model peptide and human serum albumin, and then testing the products on human immature dendritic cells and human T cells, Dietz et al. showed that "*Modification of proteins by reactive small chemicals is a key step in the activation of chemical-specific T cells in allergic contact dermatitis (ACD).*" Mass spectrometry "*revealed thiol-dependent*

peptide adduct formation at all pH values found in human skin layers" (PMID: 20631061). This is the direct chemical demonstration that the MIE is cysteine-thiol-preferring and operates across the physiological pH range of human skin (Finding F001).

Lysine-versus-cysteine reactivity partly distinguishes sensitizer classes. Using Direct Peptide Reactivity Assay data for 200 chemicals, Krutz et al. confirmed that "*The first key event in the development of either skin or respiratory sensitization is the formation of stable adducts of the chemical with host proteins,*" and found that "*Respiratory sensitizers (n = 15) showed a significant (3-fold) higher lysine reactivity than skin sensitizers (n = 129)*" (PMID: 34003265) — a difference driven largely by acid anhydrides. This establishes a residue-preference axis (cysteine for skin, relatively more lysine for respiratory) at the chemistry level (Finding F002). *System: in chemico model peptides.*

KE1 is rate-limiting, and potency is governed by the rate of cysteine-adduct formation — not a binary. This is the single most important revision to the AOP "as written." In the kinetic DPRA study of 180 chemicals correlated to mouse LLNA potency, Natsch et al. state: "*The reaction of sensitizing molecules with skin proteins is the molecular initiating event and appears to be rate-limiting, as chemical reactivity strongly correlates with sensitizer potency,*" and that "*A threshold of log kmax −2 offers a balanced accuracy of 85% for predicting GHS 1A sensitizers according to the local lymph node assay*" (PMID: 32840629). Wareing/Urbisch et al. reinforce that "*since the protein-adduct formation is determining the dose of the allergen in the skin, peptide reactivity was used to assess the potency,*" and that "*The analysis of the Cys-adducts was already sufficient; additional analysis of Lys-adducts did not improve the predictivity*" (PMID: 28882705). Roberts & Aptula's SNAr electrophile study provides a linear free-energy relationship linking Cys-peptide rate constants to LLNA potency (PMID: 24397518). Together these make KE1 a continuous, dose-determining kinetic step (Finding F012). *System: in chemico kinetics correlated to mouse LLNA potency.*

KE2 — Keratinocyte activation: electrophile sensing plus danger signaling

Sensitizers specifically activate the keratinocyte KEAP1–NRF2–ARE pathway. Testing four human keratinocyte models (primary KC, N/TERT monolayer, and Leiden and N/TERT reconstructed epidermal models) with the sensitizer DNCB versus the irritant SDS at non-toxic concentrations, Alloul-Ramdhani et al. showed "*the Keap1-Nrf2-ARE pathway is activated by the sensitizer DNCB in monolayer keratinocytes and as well as the LEMs and NEMs and not by the irritant SDS,*" and emphasize that "*In human skin, keratinocytes (KCs) are abundantly present and are key players in initiation of allergic contact dermatitis*" (PMID: 24794257). This

establishes keratinocytes as active electrophile sensors and distinguishes sensitizer from irritant at the KE2 level (Finding F003). *System: human keratinocytes / reconstructed human epidermis.*

Sensitizers activate the keratinocyte NLRP3/ASC/caspase-1 inflammasome, processing IL-1β and IL-18. Watanabe et al. demonstrated that "*key components of the inflammasome are present in human keratinocytes and that CS like trinitro-chlorobenzene induce caspase-1/ASC dependent IL-1beta and IL-18 processing and secretion,*" and that genetically "*ASC- and NALP3-deficient mice display an impaired response to CS*" (PMID: 17429439). The accompanying commentary by Yazdi et al. frames these cytokines as "*a prerequisite for the recruitment of cells into the skin that demand pro-inflammatory stimuli, such as IL-1beta and IL-18*" (PMID: 17632566). This supplies the danger-signal arm of KE2 that licenses DC priming (Finding F009). *System: human keratinocytes in vitro + NLRP3/ASC-deficient mice in vivo.*

KE3 — Dendritic cell activation: p38-MAPK, costimulation, antigen handling, and an AhR brake

DC maturation by contact sensitizers requires p38 MAPK and upregulates CD80/CD83/CD86. In human blood monocyte-derived dendritic cells, Arrighi et al. showed "*DC maturation induced by the contact sensitizers 2,4-dinitrofluorobenzene and NiSO(4), as seen by the up-regulation of CD80, CD86, and CD83, was also coupled to the phosphorylation of p38 MAPK, and was inhibited by SB203580,*" whereas "*The irritants SDS and benzalkonium chloride that do not induce DC maturation did not trigger p38 MAPK phosphorylation*" (PMID: 11238627). The irritant control establishes specificity — this is not generic cytotoxicity (Finding F004). *System: human blood-derived DC.*

Sensitizers drive CD86 upregulation plus MHC class II internalization, and AhR is a negative regulator. In human THP-1 cells, Ashikaga et al. found sensitizers "*enhanced CD86 expression on THP-1 cells, while nickel sulfate, cobalt sulfate and irritants ... did not augment CD86 expression,*" and that "*internalization of MHC class II molecules was observed when the cells were treated with some of sensitizers*" (PMID: 12423654). Using CRISPR AhR-knockout THP-1 in coculture with HaCaT keratinocytes, Sonnenburg et al. showed "*AhR-knockout THP-1 ... showed increased CD54 expression compared to wild type cells after stimulation with 10 µmol/L dinitrochlorobenzene (DNCB),*" with no such increase after the irritant SLS (PMID: 37147507) — identifying AhR as a brake on sensitizer-induced APC activation (Finding F015). *System: human monocytic cell lines (THP-1/HaCaT coculture).*

KE4 — T-cell proliferation: hapten-specific priming, MHC-I/CD8 effectors, and Treg restraint

Hapten-modified human DCs prime proliferating DNP/TNP-specific CD4⁺ and CD8⁺ T cells. Dietz et al. showed "*DNBS- but not DNCB-modified human immature dendritic cells (iDC) induced in vitro primary human T-cell responses as did 2,4,6-trinitrobenzenesulfonic acid-modified iDC as measured by dinitrophenyl (DNP)- and trinitrophenyl (TNP)-specific T-cell proliferation and interferon gamma (IFN-γ) production in CD4(+) and CD8(+) T-cell subsets*" (PMID: 20631061) — direct human evidence of hapten-specific proliferation in both subsets (Finding F005). *System: human iDC + human T cells.*

The effector program is MHC class I presentation of hapten-modified neoantigens to polyclonal CD8⁺ Tc1 cells, with CD4⁺ T cells down-regulating the response. Martin et al. reported "*the rapid induction of CD8+ Tc1 effector cells correlates with an unusually high frequency of polyclonal TNP-specific CD8+ effector T cells with specificities for a variety of MHC class I binding TNP-peptides carrying the hapten in different positions*" (PMID: 12631250). Krasteva et al. established the division of labor: the "*contact hypersensitivity (CHS) reaction to DNFB … is mediated by CD8+ effector T cells and down-regulated by CD4+ T cells*" (PMID: 11809589) (Finding F016). *System: mouse CHS for the MHC-I/CD8 effector mechanism; human in vitro for priming of both subsets.*

A hapten-specific FoxP3⁺ Treg population restrains the CD8 effector arm. Vocanson et al. showed "*2,4-Dinitrofluorobenzene immunization induced a population of CD4(+)CD25(+) Treg cells that controlled CD8(+) T-cell effector responses in a hapten-specific manner in vivo*" (PMID: 20624644) — establishing Treg tolerance as an intrinsic modulator of KE4 (Finding F005). *System: mouse CHS to DNFB.*

AO / KE827 — The adverse outcome is threshold-governed by dose per unit area

Kimber et al. established that "*it is the dose per unit area of chemical that has over-riding impact on the effectiveness of sensitization. The exception to this rule is when the area of the application site drops below a certain critical level*" (PMID: 18423821). Human experimental data from Friedmann show "*as sensitizing dose per unit area (mug cm(-2)) is increased on a log scale, there is a sigmoid dose-response curve for subsequent reactivity,*" with "*a clear inverse correlation between the strength of sensitization and the subsequent dose of antigen to which an individual will respond*" (PMID: 17854376). Basketter et al. anchor LLNA EC3 values to human induction

thresholds for 26 chemicals (PMID: 16283904). The outcome is therefore quantitative, threshold-bearing, and governed by μg/cm² (Finding F013). *System: human experimental sensitization + mouse LLNA anchoring.*

Alternative and non-covalent initiating routes

Pre-haptens and pro-haptens act upstream of KE1, not around it. Hennen et al. (HaCaT/THP-1 coculture) state that many sensitizers "*require activation outside or inside the skin by nonenzymatic oxidation (prehaptens) or metabolic transformation (prohaptens) prior to being able to induce an immune response,*" and showed mechanistically that "*Inhibition of cytochrome P450 or NAD(P)H:quinone oxidoreductase (NQO1) activity reduced the prohapten-mediated upregulation of CD86 on cocultured THP-1 cells*" (PMID: 21742781). Clinically, Hagvall et al. confirm "*Geraniol is a pro-hapten and a pre-hapten, and sensitizers are formed in the autoxidation and skin metabolism of geraniol*" (PMID: 23510343). Both routes generate the reactive electrophile and feed into the covalent MIE — they are **inside AOP 40** (Finding F006). *System: human cell coculture + clinical patch-test cohort.*

Metal sensitizers coordinate human TLR4 histidines — outside the covalent MIE. Schmidt et al. showed "*Ni(2+) triggered an inflammatory response by directly activating human Toll-like receptor 4 (TLR4),*" that "*the non-conserved histidines 456 and 458 of human TLR4 are required for activation by Ni(2+) but not by the natural ligand lipopolysaccharide,*" and that "*Ni(2+)-induced TLR4 activation was species-specific, as mouse TLR4 could not generate this response*" (PMID: 20711192). Oblak et al. extended this to cobalt: "*Activation of TLR4 by cobalt required MD-2 and was abolished by human TLR4 mutations of histidine residues at positions 456 and 458*" (PMID: 25803856). Zoroddu et al. provide the biophysical binding detail (Ni(II) binds H431/H456/H458 of the hTLR4 429–460 peptide) (PMID: 24169691). This non-covalent coordination route lies **outside AOP 40 as written** (Finding F007). *System: human cells + humanized transgenic mice; biophysics on peptide fragment.*

AOP revision: Langerhans cells are dispensable and regulatory

Kaplan et al., using Langerin-DTA transgenic mice with constitutive LC absence, found "*Unexpectedly … contact hypersensitivity (CHS) was amplified rather than abrogated in the absence of LCs,*" and that "*LCs not only were dispensable for CHS, but they served to regulate the response, a previously unappreciated function,*" acting "*during the priming and not the effector phase*" (PMID: 16356859). The review-level synthesis by Gober & Gaspari adds the unresolved

gap: "*The identity of the antigen presenting cells that complement Langerhans cells has yet to be identified*" (PMID: 18460878). This revises the APC step of the AOP (Finding F008). *System: mouse genetic ablation.*

Interindividual variation

Demonstrated mechanisms (dose-tuning). Filaggrin loss-of-function: Novak et al. (n=1502 KORA cohort) report "*Impaired skin barrier function facilitates the penetration of contact allergens and irritants into the epidermal layer and is regarded as an important cofactor,*" with FLG R501X/2282del4 showing "*an association with contact sensitization to nickel ... but not with other contact allergens*" (PMID: 18049447); Thyssen et al. confirm "*increased penetration of allergens and chemicals in filaggrin-deficient skin*" (PMID: 23301728). Biotransformation: Kezic reports "*polymorphisms in genes encoding N-acetyltransferases were shown to modify the risk for sensitization to p-phenylenediamine*" (PMID: 21329569) (Finding F011). Microbiome: Shang et al. (reconstructed human skin) found "*S. mitis increased interleukin (IL)-6, CXCL8, CCL2, CCL5, and CCL20 secretion in RHS ... co-application with nickel further increased cytokine secretion,*" indicating "*the important influence of commensal microbes and the route of exposure on the host's response to metals*" (PMID: 32677222) (Finding F014). Treg tolerance: Cavani et al. (6 healthy nonallergic human donors) showed nickel responsiveness "*was strongly augmented (mean increment ± SD, 240 ± 60%) when cells were depleted of CD25(+) Treg,*" and that "*CD25(+) Treg strongly and dose-dependently inhibited nickel-specific activation of CD25(−) T lymphocytes ... in a cytokine-independent, but cell-to-cell contact-dependent, manner*" (PMID: 14634084) — a *demonstrated human* tolerance mechanism (Finding F017).

Associations only — HLA for defined contact haptens. Sources disagree. Silvennoinen-Kassinen et al. (1997, 55 nickel-sensitive vs 54 controls) report "*The allele and phenotype frequencies of TAP2B were significantly (p = 0.019 and 0.012, respectively) increased in nickel-sensitive subjects versus controls*" (PMID: 9303338), yet a separate study is titled "*No significant association between HLA and nickel contact sensitivity*" (PMID: 12731579). The strong HLA case is for a *protein/respiratory* allergen, not a contact hapten: Jeal et al. found "*HLA phenotype is an important determinant of individual susceptibility to sensitization and asthma among laboratory animal workers*" (HLA-DR7, OR 1.82) (PMID: 12704360). For contact haptens, HLA remains a weak, inconsistent association, not a demonstrated restriction mechanism (Finding F010).

Mechanistic Model / Interpretation

The decade of work resolves AOP 40 into discrete causal steps with named participants. The causal chain, with revisions flagged, is:

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PRE-HAPTEN (abiotic O2 oxidation) ─┐
PRO-HAPTEN (KC CYP450 / NQO1) ─────┤ [inside AOP40, upstream of KE1]
DIRECT ELECTROPHILE ───────────────┘
          │
          ▼
  KE1 / MIE  Covalent adduct on protein CYS thiol (>LYS)
             Chemistry: Michael addition / SNAr / Schiff base / acyl transfer
             *** RATE of cys-adduct formation (log kmax) is RATE-LIMITING
                 and sets POTENCY — continuous, not binary ***
          │
          ▼
  KE2  Keratinocyte activation (active sensor, not bystander)
       • KEAP1–NRF2–ARE electrophile sensing (sensitizer-specific vs SDS)
       • NLRP3 / ASC / caspase-1 inflammasome → IL-1β, IL-18 (danger signal)
          │
          ▼
  KE3  Dendritic cell maturation
       • p38-MAPK → CD80 / CD83 / CD86 up; MHC-II internalization
       • AhR = negative regulator (brake)
          │
          ▼
  KE4  Hapten-specific T-cell proliferation
       • MHC class I presentation of hapten-modified NEOANTIGEN peptides
       • polyclonal CD8+ Tc1 effectors (IFN-γ) = drivers of CHS
       • CD4+ T cells down-regulate; FoxP3+/CD25+ Treg restrain (tolerance)
       *** Priming APC = migratory/dermal DC, NOT obligatorily Langerhans cells
           LCs dispensable & REGULATORY (revision); replacing subset unknown (gap)
          │
          ▼
  AO (KE827)  Skin sensitization
       *** Threshold-bearing, SIGMOID dose–response governed by
           DOSE PER UNIT AREA (µg/cm²) ***

PARALLEL ROUTE OUTSIDE AOP40 — METALS:
  Ni2+ / Co2+ coordinate non-conserved HIS456/HIS458 of human TLR4 (+MD-2)
  → direct innate inflammation; human-specific (mouse TLR4 inert)
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The unifying theme is **quantitation**. The AOP "as written" reads as a qualitative sequence of binary key events, but the mechanistic literature shows that the chain is governed throughout by two quantities: at the chemistry end, the *rate constant* of cysteine adduct formation sets

potency (KE1), and at the biology end, *dose per unit area* sets whether induction crosses threshold (AO). Interindividual modifiers — filaggrin barrier state, NAT detoxification, microbiome, Treg tone — operate predominantly by adjusting how much reactive hapten reaches viable epidermis or how strongly the innate/adaptive arms respond to it, i.e. they tune these same quantities rather than introducing new key events.

Key event	Named participants	Cause → consequence	Species/system
KE1/MIE	Cys thiol, Lys (secondary); HSA; electrophile	Covalent adduct forms; rate sets potency	in chemico; human DC/T
KE2	KEAP1, NRF2, ARE; NLRP3, ASC, caspase-1; IL-1β, IL-18	Electrophile sensing + inflammasome danger signal	human KC / RHE; KO mice
KE3	p38-MAPK; CD80/83/86; MHC-II; AhR (brake)	Costimulation up, antigen internalized	human DC / THP-1
KE4	MHC-I; $CD8^+$ Tc1 (IFN-γ); $CD4^+$; $FoxP3^+/CD25^+$ Treg	Neoantigen presentation primes effectors; Treg restrains	mouse CHS; human in vitro
AO	dose/area (μg/cm²)	Sigmoid, threshold induction	human + mouse LLNA
Metals (outside)	Ni^{2+}/Co^{2+}, human TLR4 H456/H458, MD-2	Non-covalent coordination → innate inflammation	human cells; humanized mice

Evidence Base

PMID	Study (abbrev.)	Role	System
20631061	Dietz 2010	Supports KE1 thiol-dependence + KE4 human priming	in chemico + human cells
34003265	Krutz 2021	Lys vs Cys sensitizer-class split	in chemico
32840629	Natsch 2020 (kDPRA)	KE1 rate-limiting; potency = rate	in chemico vs mouse LLNA
28882705	Urbisch/Wareing 2017	Cys-adduct = internal dose; Cys sufficient	in chemico
24397518	Roberts & Aptula 2014	SNAr LFER rate↔potency	in chemico/mouse
24794257	Alloul-Ramdhani 2014	KE2 NRF2 sensitizer-specific	human KC/RHE
17429439	Watanabe 2007	KE2 inflammasome	human KC + KO mice
17632566	Yazdi 2007	Inflammasome commentary	synthesis
11238627	Arrighi 2001	KE3 p38-MAPK/CD80-83-86	human DC
12423654	Ashikaga 2002	KE3 CD86 + MHC-II internalization	human THP-1
37147507	Sonnenburg 2023	AhR brake on KE3	human THP-1/ HaCaT
12631250	Martin 2003	KE4 MHC-I/CD8 Tc1 neoantigen	mouse
11809589	Krasteva 2002	CD8 effector / CD4 down-regulate	mouse
20624644	Vocanson 2010	Hapten-specific Treg restrains CD8	mouse
18423821	Kimber 2008	AO = dose/area	human/mouse
17854376	Friedmann 2007	Sigmoid human dose–response	human
16283904	Basketter 2005	EC3 ↔ human threshold	human/mouse
21742781	Hennen 2011	Pre/pro-hapten activation	human coculture
23510343	Hagvall 2013	Geraniol pre+pro-hapten clinical	human cohort

PMID	Study (abbrev.)	Role	System
20711192	Schmidt 2010	Ni^{2+} → human TLR4 H456/458	human + humanized mice
25803856	Oblak 2015	Co^{2+} TLR4/MD-2	human
24169691	Zoroddu 2014	Ni(II)–hTLR4 binding biophysics	peptide
16356859	Kaplan 2005	LCs dispensable/regulatory	mouse ablation
18460878	Gober & Gaspari 2008	APC gap	synthesis
18049447	Novak 2008	FLG→nickel penetration	human cohort n=1502
23301728	Thyssen 2013	FLG-deficient penetration	human mechanistic
21329569	Kezic 2011	NAT→PPD; low predictive value	human synthesis
32677222	Shang 2020	Microbiome amplifies Ni response	human RHS
14634084	Cavani 2003	Human $CD25^+$ Treg nickel tolerance	human primary T
9303338	Silvennoinen-Kassinen 1997	TAP2B nickel association (+)	human n≈109
12731579	Silvennoinen-Kassinen	No HLA–nickel association (−)	human
12704360	Jeal 2003	HLA-DR7 ↔ protein allergen (contrast)	human

Noted disagreements reported, not silently resolved: (i) HLA–nickel association is positive in one small cohort (PMID: 9303338) and null in another (PMID: 12731579); neither is a large modern GWAS, so the question is unresolved, and the robust HLA signal exists only for protein/respiratory allergens (PMID: 12704360). (ii) For cobalt-alloy debris, danger signaling (NLRP3) may dominate over direct TLR4 activation (PMID: 27467577), a caveat on the metal-TLR4 route in particulate contexts.

Limitations and Knowledge Gaps

1. **Species mismatch in the effector arm.** The cleanest MHC-I/$CD8^+$ Tc1 neoantigen mechanism (Martin 2003; Krasteva 2002; Vocanson 2010) is **mouse** CHS. Human data confirm priming of both $CD4^+$ and $CD8^+$ subsets (Dietz 2010) but do not independently establish the MHC-I-restricted polyclonal CD8 effector dominance in humans. The CD8- effector assignment is therefore a strong mouse inference, not a fully human claim.
2. **The specific in-vivo protein target is unknown.** Current in chemico assays (DPRA/kDPRA) quantify **bulk cysteine-peptide depletion** as a surrogate for haptenation. No retrieved primary source identifies the specific epidermal protein(s) whose modification actually drives KE2/KE3 in vivo. Whether "covalent binding to a specific protein target" versus "bulk peptide depletion" drives the next event remains open (Finding F018 synthesis).
3. **The priming APC is unidentified.** With LCs shown dispensable/regulatory, which migratory/dermal DC subset primes the T-cell response is explicitly unresolved (PMID: 18460878).
4. **Host-susceptibility genetics lacks predictive power.** Kezic states "*the predictive value of these genetic markers is too low for the reasonable selection of susceptible individuals*" (PMID: 21329569). FLG effects are nickel-specific; NAT effects are PPD-specific; HLA for contact haptens is inconsistent.
5. **Metal-TLR4 route is human-specific and context-dependent.** It is human-specific (mouse TLR4 inert), limiting animal modeling, and in particulate/implant contexts danger signaling may dominate (PMID: 27467577).
6. **Microbiome and AhR evidence is model-based.** The S. mitis and AhR findings derive from reconstructed-skin and cell-line coculture systems, not in-vivo human skin.

Proposed Follow-up Experiments / Actions

1. **Identify the in-vivo haptenated protein target.** Perform adductomic mass spectrometry on viable human epidermis (ex vivo skin or reconstructed human epidermis) exposed to a strong sensitizer at realistic $\mu g/cm^2$, to move from bulk cysteine depletion to a named protein/residue target and test whether modification of a specific carrier drives KE2/KE3. This directly addresses Gap 2.

2. **Resolve the human priming APC.** Use single-cell profiling and selective depletion of candidate dermal/migratory DC subsets in human skin explants or humanized models to identify which APC replaces the dispensable Langerhans cell (Gap 3).
3. **Confirm the human CD8$^+$ Tc1 effector mechanism.** Isolate hapten-specific CD8$^+$ T cells from sensitized human donors and verify MHC class I-restricted, hapten-modified neoantigen recognition and IFN-γ Tc1 polarity, to convert the mouse-derived effector model into a human claim (Gap 1).
4. **Adequately power the HLA question.** A large, modern, well-controlled HLA/GWAS study of defined contact haptens (nickel, PPD) is needed to settle the contradictory small-cohort associations; prior studies are n≈50–110 and conflict.
5. **Integrate kinetics into quantitative risk assessment.** Given KE1 rate-limitation, systematically pair log kmax (kDPRA) with dose-per-area exposure metrics to build a quantitative kinetics-to-threshold model linking the chemistry endpoint to the human AO.
6. **Validate microbiome and AhR modulation in vivo.** Test whether commensal manipulation or AhR agonism measurably shifts human sensitization thresholds, to establish these from plausibility (reconstructed-skin/cell-line) to demonstrated human modifiers.

Report compiled from 18 confirmed findings across 5 iterations and 64 papers reviewed. Every mechanistic claim above carries a PMID and a verbatim quote from the cited work; species and system are stated for each. Mouse results are labeled as such and not presented as human claims.
